

A Brief Overview of “Big Tent” Inquiry-Based Learning

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Thank you Tara, Grigorii, Boris!

Outline

1. Introduction and setting the stage
2. 4 pillars of IBL
3. Classic/core example of IBL
4. (Brief) overview of the education literature
5. Adaptations we can use here at UofT

Introduction, Setting the Stage

- IBL = Inquiry Based Learning
- Context matters: teaching methodologies develop within a context (students, course topics, instructor, institution, class sizes, local society,...)
- Different people define active learning, IBL in different ways. These definitions have been used in different places and over time. (Analogy: birds that have adapted to different environments.)

Education is a system and education is a cultural activity.

Education is not a zero sum game.

Education is not binary as in IBL = on/off. (Think of dials on a guitar amplifier.)



You're peculiar!

1. Only a small percentage of university graduates go on to earn advanced degrees in math.
2. You're an outlier, like Michael Jordan is an outlier.
3. "I like lectures..."
4. "It worked for me..."
5. As an expert, you are far from the beginning learner.



Acknowledging the past

The IBL movement has intersections with the Moore Method. R. L. Moore, while a great educator, was a problematic person due his views on race, gender.

Moore denied black students and women from taking his classes, and would be at registration and controlled who could be in his classes.

Ironically, his teaching methods in his class were more egalitarian, and today modernized variants of IBL are equity supporting.

Also it's important to keep in mind that [...]isms are systems.

There was a movement about 12 years ago to rebrand to IBL.

“Big Tent” a modernized, research-based evolution

“Big Tent” IBL is a broad framework of approaches that can be adapted to many situations. The evolution from the early days was influenced by education researchers and people who cares about fairness, equity, and inclusion.

The Four Pillars of IBL Teaching

1. Deep engagement in rich mathematics
2. Regular opportunities to collaborate with students and instructor
3. Instructor inquiry into student thinking
4. Instructor focus on equity

Guided, scaffolded, tailored to students' need. IBL is not open, unsupported discovery.

(Laursen, Rasmussen 2019, *I on the Prize: Inquiry Approaches in Undergraduate Mathematics*)

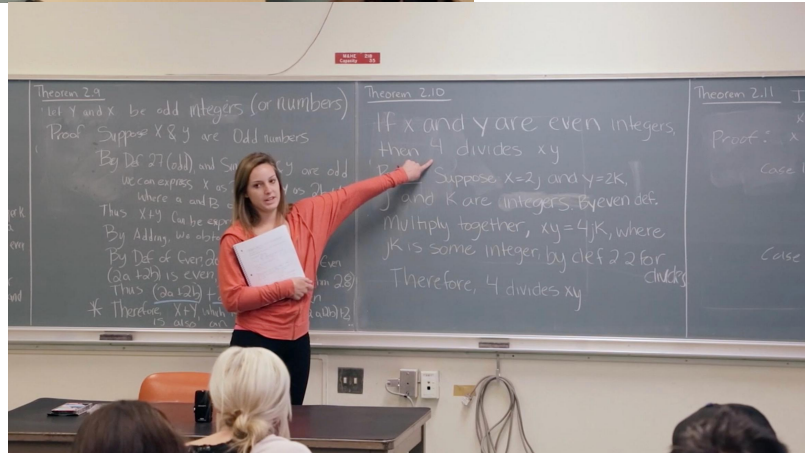
A “standard” example of IBL as a historical starting point



Prof. Mike Starbird, Univ. of Texas, 2003



Cal Poly San Luis Obispo



Key features

1. Carefully sequences (scaffolded) of problems that lead to major theorems/ideas in a course broken into appropriate sized steps.
2. Students are asked to try problems without looking up solutions.
3. Students present their solutions/proofs in class, shifting focus to student thinking. Math authority is shifted to logic and reasoning (not the teacher or book).
4. Student with the fewest number of presentations has priority.
5. After a peer-review of the problem is done, the class moves on to the next problem or if done then working on the next round of problems.
6. Instructor's role shifts to a facilitation role.
7. *not all content has to be covered this way and there are ways to address coverage via mini lectures, reading assignments.

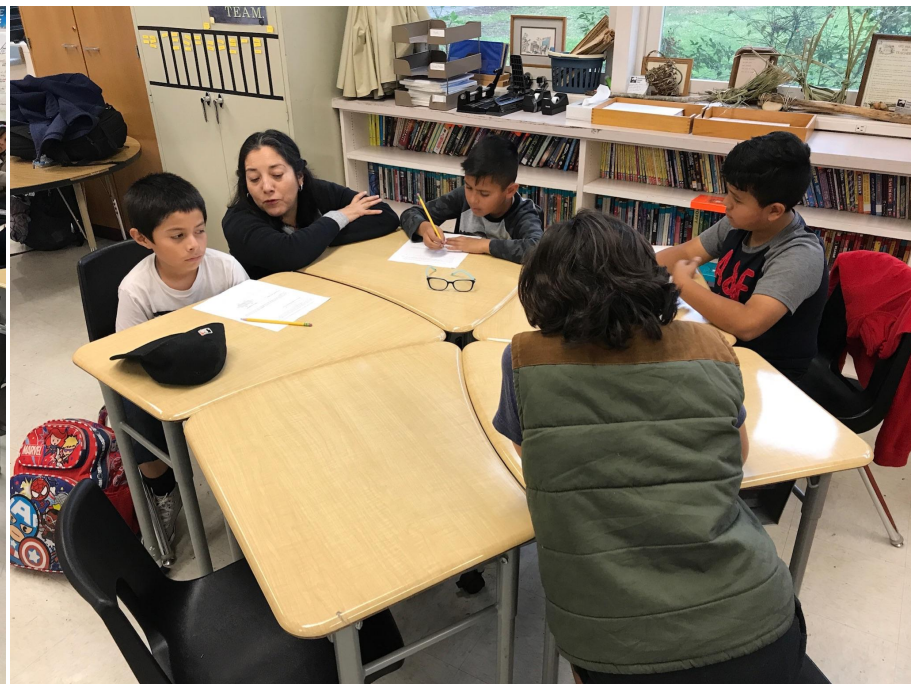
Other examples

Mike Starbird 10 years later (groups and presentations)





First/second year courses with more reliance on group tasks



Elementary school setting

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*Listening to your students' thinking is so important

Highlights from the Math Ed Literature

- TL;DR Passive lectures is not fully effective and evidence supports strongly the use of well-implemented active learning/IBL.
- Goal is like in medical fields to update our practices as we receive new knowledge.
- Reminder that you're peculiar
- No guilt/shame
- Brief overview

What does Math Ed research say about students' ability to ...?

- Unpack mathematical statements?
- Problem solve?
- Apply their knowledge even if they have requisite skills?

Evidence suggests students are not usually successful at these things, b/c they are not provided opportunities to learn how to think mathematically.

Schoenfeld 1988, Muis 2004, Selden and Selden 1995, 1999, 2003, Dreyfus 1999, Sowder and Harel 2003, Weber, Alcock and Webber 2004, Tall 1994

Nonavailing beliefs are beliefs that hinder learning

K. Muis 2004 (metastudy)

1. Memorizing facts and formulas and practicing procedures are sufficient to learn mathematics.
2. Mathematics textbook problems can only be solved using the methods described in the textbook.
3. Teachers and textbooks are the mathematical authorities.
4. School mathematics is driven by rules and memorization, and is driven by procedures rather than concepts.
5. If a problem takes longer than 5-10 minutes, then there is something wrong with the student or the problem.
6. The goal of mathematics is to obtain one correct answer and do it quickly.
7. The teacher is the only source of determining whether an answer is correct or incorrect.
8. Students' role in the classroom is to receive knowledge by paying attention in class and to demonstrate it has been received by producing right answers.
9. The teacher's role is to transmit knowledge and verify that it has been transmitted.
10. Only geniuses have what it takes to be good at mathematics.
11.

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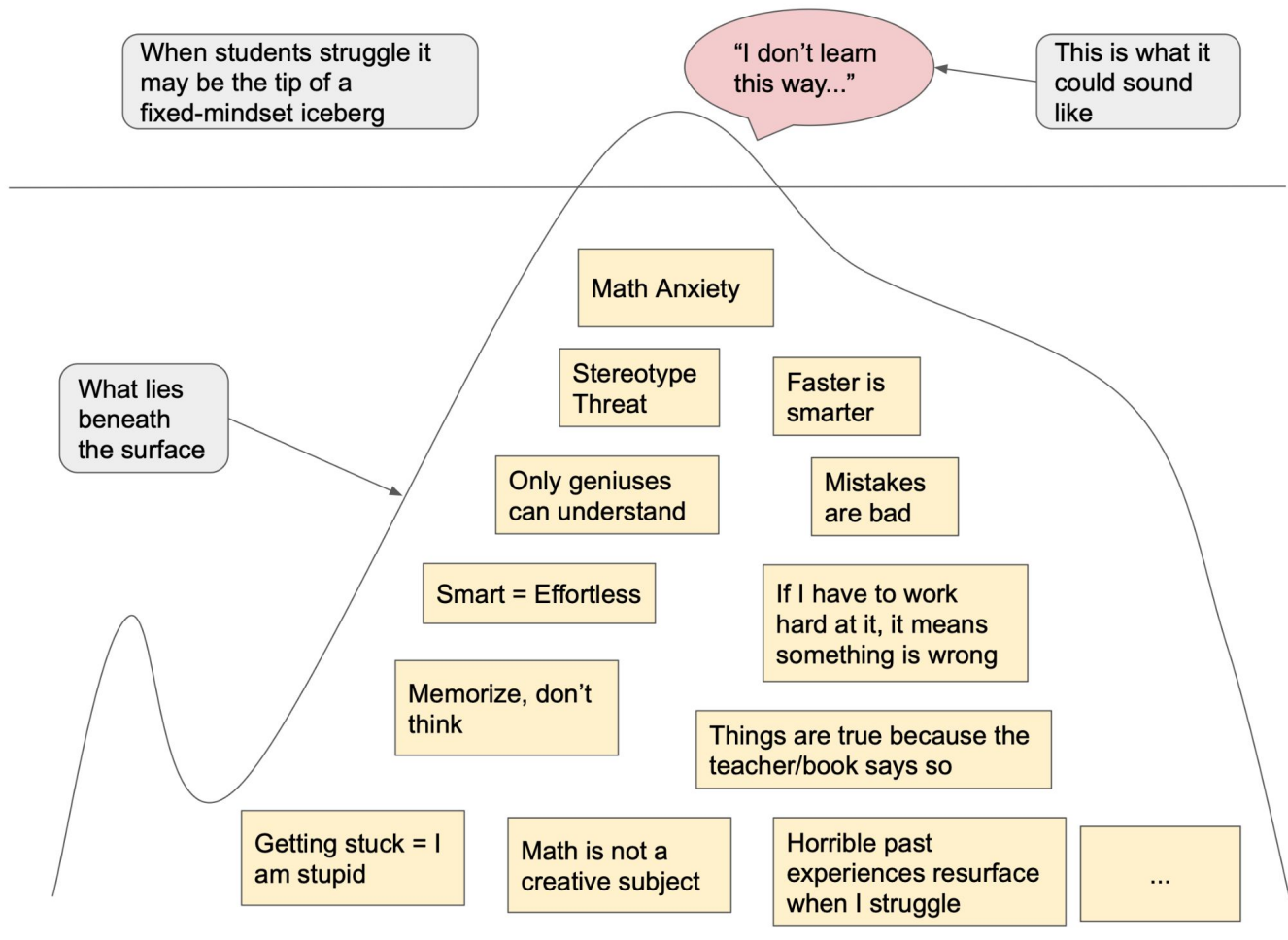
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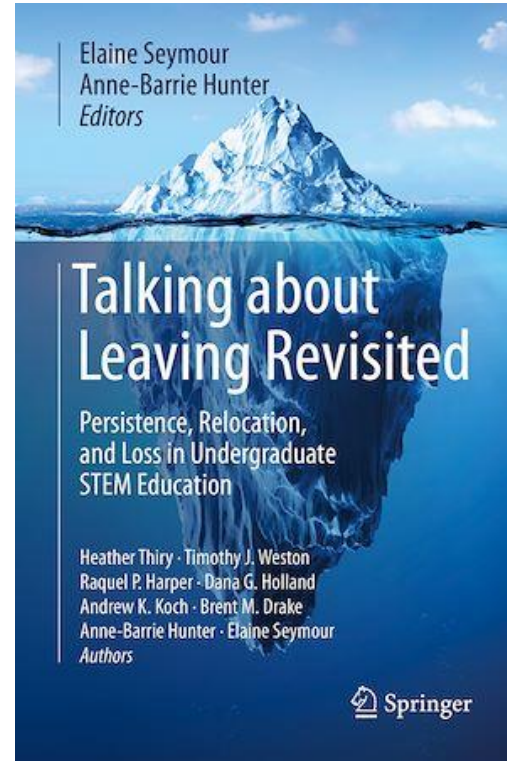
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Math Anxiety Iceberg Diagram



Students leaving STEM

1. 50% of STEM majors switch (US Data)
2. Teaching is the main reason
3. **Good students leave, too.** Cannot predict leaving based on things like GPA.



Equity

Women 1.5 times more likely to leave STEM after Calculus

Ellis, J., Fosdick, B. K., & Rasmussen, C. (2016). Women 1.5 Times More Likely to Leave STEM Pipeline after Calculus Compared to Men: Lack of Mathematical Confidence a Potential Culprit. PloS One, 11(7), e0157447–e0157447.

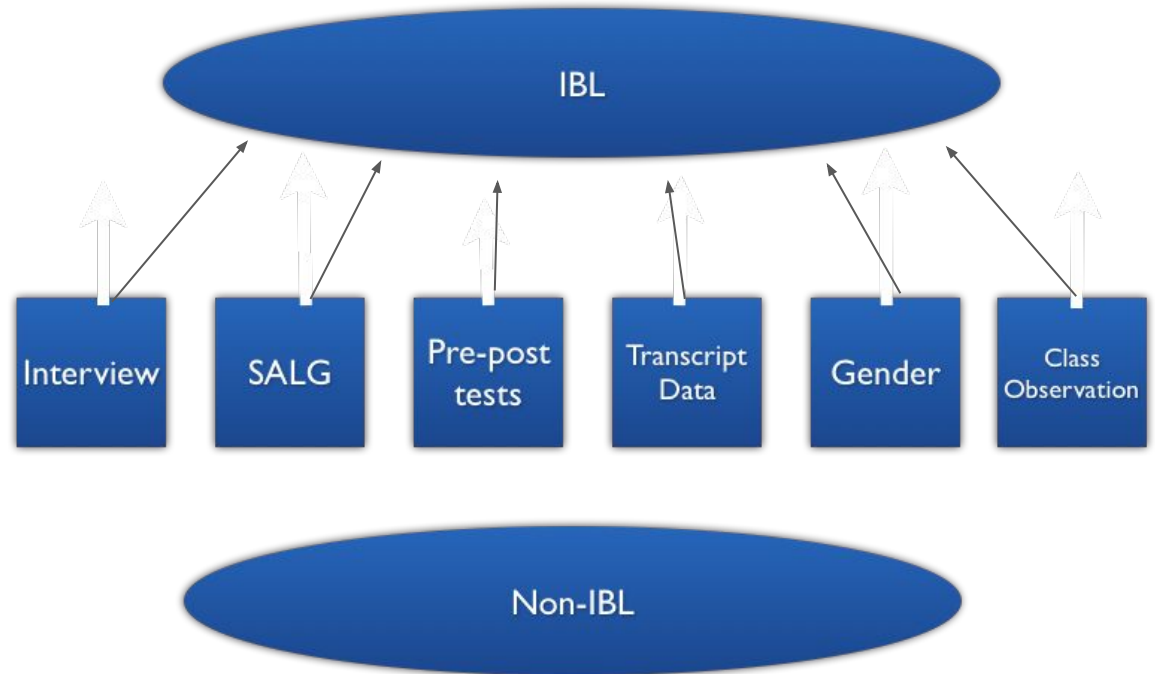
<https://doi.org/10.1371/journal.pone.0157447>

Evidence for IBL

IBL Study by Sandra Laursen et al.

1. All students benefit.
2. Lower achieving students benefit significantly.
3. Gender differences reduced/eliminated in IBL courses vs. non-IBL

Colorado Study



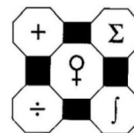
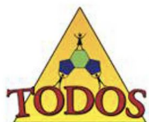
Freeman et al (2014)

“If the experiments analyzed here had been conducted as randomized controlled trials of medical interventions, they may have been stopped for benefit—meaning that enrolling patients in the control condition might be discontinued because the treatment being tested was clearly more beneficial.”

Theobold, et al 2020

“Active learning benefits all students but offers **disproportionate benefits for individuals from underrepresented groups**. Widespread implementation of high-quality active learning can help reduce or eliminate achievement gaps in STEM courses and promote equity in higher education.”

Conference Board of the Mathematical Sciences



CBMS Active Learning Statement

“A wealth of research has provided clear evidence that active learning results in better student performance and retention than more traditional, passive forms of instruction alone. ”

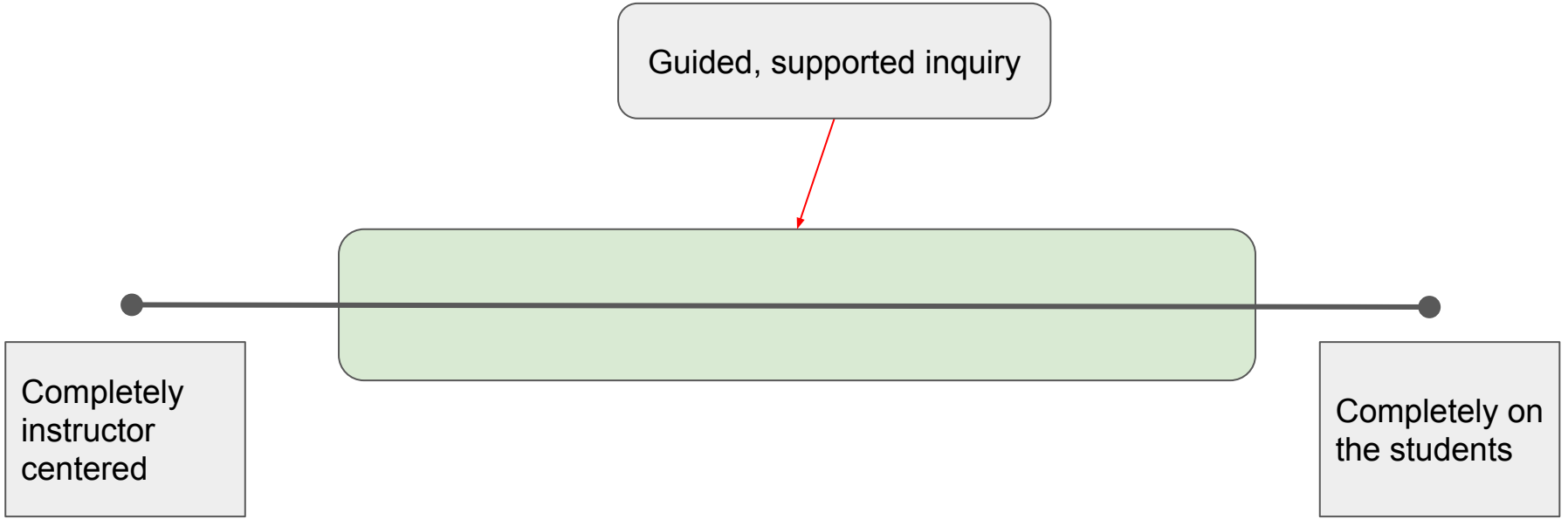
“...we call on institutions of higher education, mathematics departments and the mathematics faculty, public policy-makers, and funding agencies to invest time and resources to ensure that effective active learning is incorporated into post-secondary mathematics classrooms.”

Equity is not automatic

Group work can potentially be inclusive or exclusive depending on how well it is implemented.

Ernest, J. B., Reinholz, D. L., & Shah, N. (2019). Hidden competence: women's mathematical participation in public and private classroom spaces. *Educational Studies in Mathematics*, 102(2), 153–172.

<https://doi.org/10.1007/s10649-019-09910-w>



Adapting to our local context at UofT



Adaptations for large classes

Riff off of Think-Pair-Share

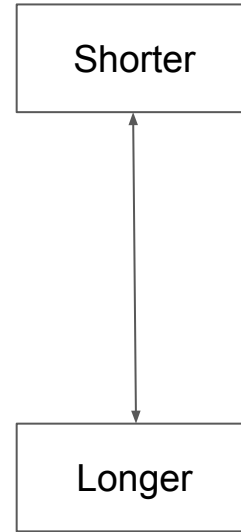
Think time (1-2 minutes) for students to process for themselves

Pair “Turn and talk to your neighbour...”

Share Make public the ideas so that all students have access to them.
Student/pair share, assign students/pairs to share, reveal, etc.

Riffs off of Think-Pair-Share

1. Think and go (instructor reveal)
2. Think, pair, go (instructor reveal)
3. Think, pair, share
4. 1-2-4-All (have groups talk to another group)



Guiding philosophy/practice

1. Keep it simple!
2. Implement simple things well.
3. Quality over quantity/fancy.
4. Post slides/notes.

IBL Handouts used in lectures

- Create a pdf with tasks written on them and send them to students before class.
- Work through these in class.
- Use a mix of small group work (pairs) and riffs on think-pair-share.

MAT 223 Module 3 and Appendix 2 Lecture Handout

1. The first concept we learn about in Module 3 is the idea of span. Read the definition.

Span. The span of a set of vectors V is the set of all linear combinations of vectors in V . That is,
 $\text{span } V = \{ \vec{v} : \vec{v} = \alpha_1 \vec{v}_1 + \alpha_2 \vec{v}_2 + \cdots + \alpha_n \vec{v}_n \text{ for some } \vec{v}_1, \vec{v}_2, \dots, \vec{v}_n \in V \text{ and scalars } \alpha_1, \alpha_2, \dots, \alpha_n \}.$

Additionally, we define $\text{span}\{\} = \{ \vec{0} \}.$

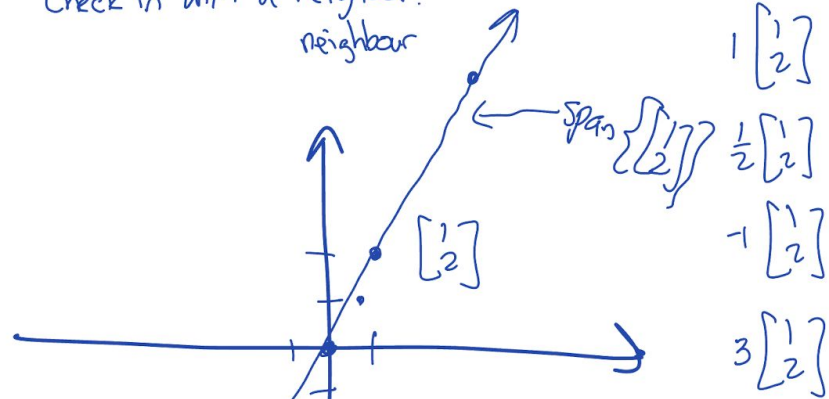
Read.

$$\text{span}\phi = \{ \vec{0} \}$$

2. The span is a set of vectors, so we can represent them with graphs. Graph $\text{span}\left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\}.$ $\mathbb{R}^2$

Think

*Check in with a neighbor.
neighbor*

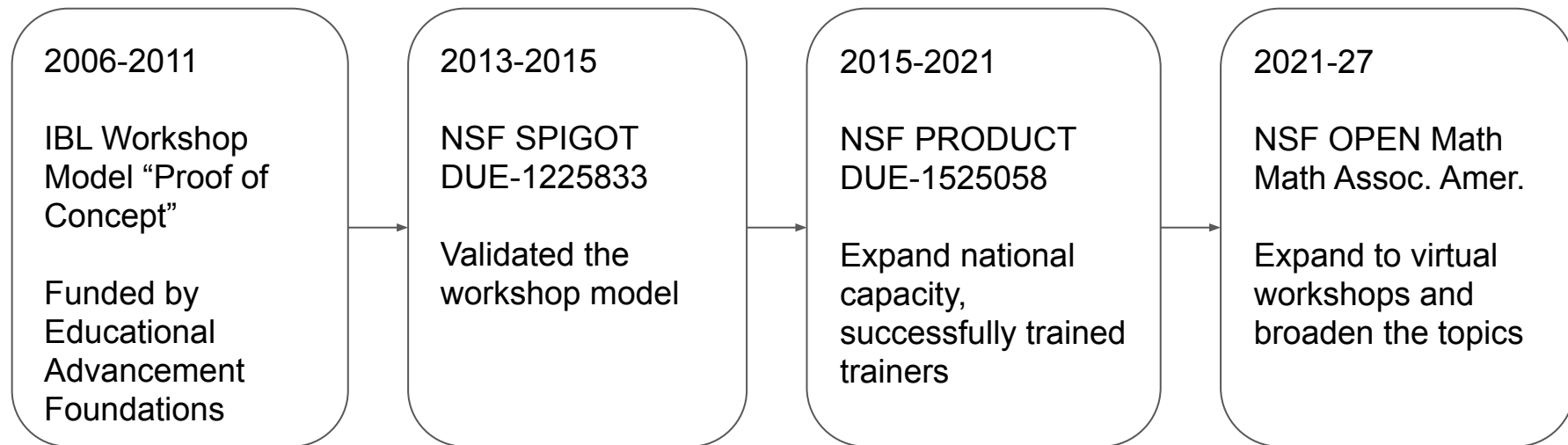


Other examples

1. Here's a definition... what are some examples and almost, but non-examples.
2. Try this example... turn and talk... "Person X has a good solution, let me share Person X's idea..."
3. Study these examples and make one or more conjectures...
4. Here's a theorem. Think about how you would prove it.
5. Here's the first part of the proof of the theorem. Try and complete the rest/next step.
6. Here's the full proof of the theorem. With your partner, identify the key steps and big ideas.
7. Now that we have Theorem X, what are possible corollaries...

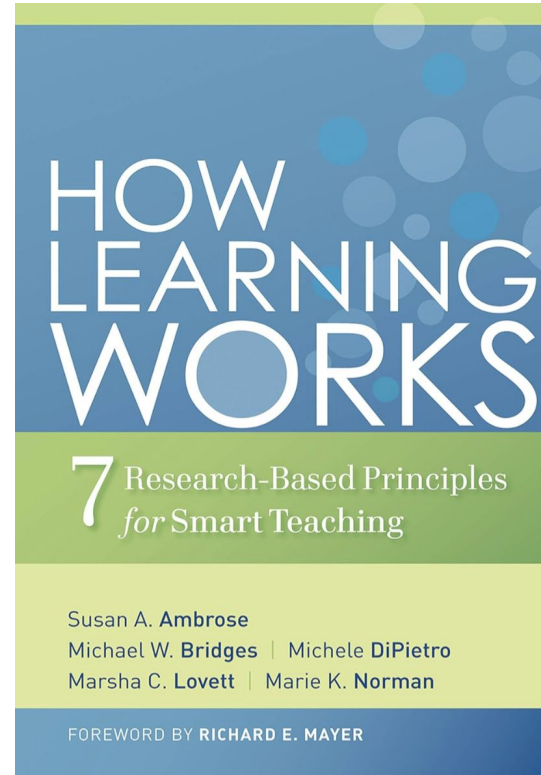


Teaching practices are learnable. Yes, you can do IBL!



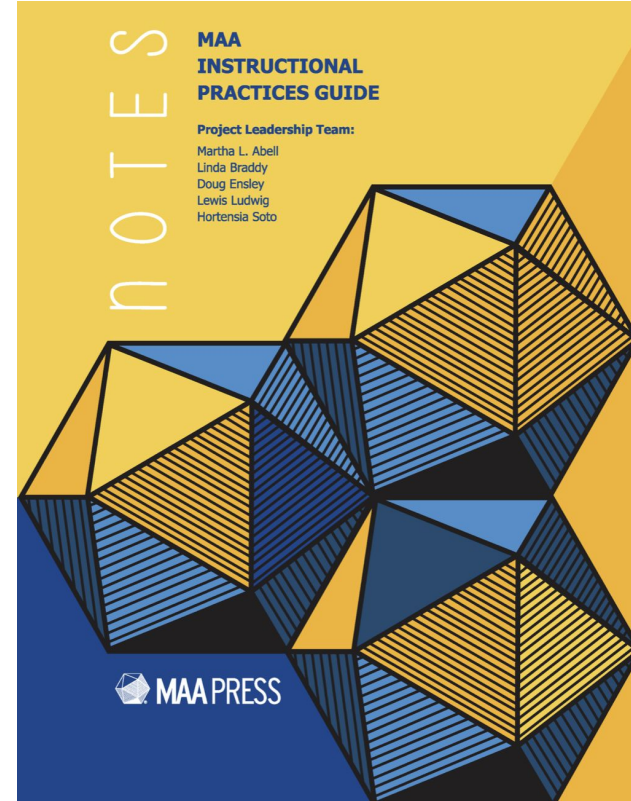
How Learning Works, Ambrose et al

Bridges the gap between research and practice for all university/college instruction. (Available at our library.)



MAA Instructional Practices Guide

Solid reference for teaching methods, practices and is a free download.



SIGMAA IBL

<http://sigmaa.maa.org/ibl/>

IBL focused **S**pecial **I**nterest **G**roup of the **MAA**.

References List

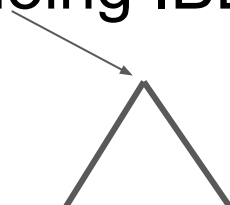
<https://tinyurl.com/TMSSY2026>

Thank you!

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Extras

Sense-Making “Continental Divide”, “When am I doing IBL?”



Non-IBL

Instructor does essentially all of the processing.

IBL Zone

Students engaged in doing math. Students do significant portion of the sense-making regularly.